

Research Notes on Regularization machine learning

Regularization machine learning

>> Section 1

Dropout In the context of neural networks, the Dropout technique repeatedly ignores random subsets of neurons during training, which simulates the training of multiple neural network architectures at once to improve generalization.

>> Section 2

Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

>> Section 3

$R(f_1 \dots f_T) = \sum_{t=1, t \neq s}^T \|f_t - f_s\|_{M_{ts}}^2$ $\{\displaystyle R(f_1 \cdots f_T) = \sum_{t,s=1, t \neq s}^T \|f_t - f_s\|_{M_{ts}}^2\}$ for a given symmetric similarity matrix M $\{\displaystyle M\}$.

>> Section 4

Group sparsity without overlaps Groups of features can be regularized by a sparsity constraint, which can be useful for expressing certain prior knowledge into an optimization problem. In the case of a linear model with non-overlapping known groups, a regularizer can be defined:

>> Section 5

$R(w) = \sum_{i=1}^n \sigma(W_i)$ $\{\displaystyle R(w) = \sum_{i=1}^n \sigma(W_i)\}$ where $\sigma(W)$ $\{\displaystyle \sigma(W)\}$ is the eigenvalues in the singular value decomposition of W $\{\displaystyle W\}$.

>> Section 6

Group sparsity with overlaps The algorithm described for group sparsity without overlaps can be applied to the case where groups do overlap, in certain situations. This will likely result in some groups with all zero elements, and other groups with some non-zero and some zero elements. If it is desired to preserve the group structure, a new regularizer can be defined:

>> Section 7

$\min_w \frac{1}{n} \|X^T w - Y\|^2 + \lambda \sum_{i=1}^n \|w_i\|_2^2$ $\{\displaystyle \min_w \frac{1}{n} \|X^T w - Y\|^2 + \lambda \sum_{i=1}^n \|w_i\|_2^2\}$

$$\|X^T w - Y\|^2 + \lambda \sum_{i=1}^n \|w_i\|_2^2$$

>> Section 8

Regularizers for sparsity Assume that a dictionary ϕ_j $\{\displaystyle \phi_j\}$ with dimension p $\{\displaystyle p\}$ is given such that a function in the function space can be expressed as:

>> Section 9

Without bounds on the complexity of the function space (formally, the reproducing kernel Hilbert space) available, a model will be learned that incurs zero loss on the surrogate empirical error. If measurements (e.g. of x_i $\{\displaystyle x_i\}$) were made with noise, this model may suffer from overfitting and display poor expected error. Regularization introduces a penalty for exploring certain regions of the function space used to build the model, which can improve generalization.

>> Section 10

where V $\{\displaystyle V\}$ is an underlying loss function that describes the cost of predicting $f(x)$ $\{\displaystyle f(x)\}$ when the label is y $\{\displaystyle y\}$, such as the square loss or hinge loss; and λ $\{\displaystyle \lambda\}$ is a parameter which controls the importance of the regularization term. $R(f)$ $\{\displaystyle R(f)\}$ is typically chosen to impose a penalty on the complexity of f $\{\displaystyle f\}$. Concrete notions of complexity used include restrictions for smoothness and bounds on the vector space norm. A theoretical justification for regularization is that it attempts to impose Occam's razor on the solution (as depicted in the figure above, where the green function, the simpler one, may be preferred). From a Bayesian point of view, many regularization techniques correspond to imposing certain prior distributions on model parameters. Regularization can serve multiple purposes, including learning simpler models, inducing models to be sparse and introducing group structure into the learning problem. The same idea arose in many fields of science. A simple form of regularization applied to integral equations (Tikhonov regularization) is essentially a trade-off between fitting the data and reducing a norm of the solution. More recently, non-linear regularization methods, including total variation regularization, have become popular.

>> Section 11

$$L_S[f_n] = \frac{1}{n} \sum_{i=1}^n V(f_n(x_i), y_i) \{\displaystyle L_S[f_n] = \frac{1}{n} \sum_{i=1}^n V(f_n(x_i), y_i)\}$$

>> Section 12

Explicit regularization is regularization whenever one explicitly adds a term to the optimization problem. These

Research Notes on Regularization machine learning

Regularization machine learning

terms could be priors, penalties, or constraints. Explicit regularization is commonly employed with ill-posed optimization problems. The regularization term, or penalty, imposes a cost on the optimization function to make the optimal solution unique. Implicit regularization is all other forms of regularization. This includes, for example, early stopping, using a robust loss function, and discarding outliers. Implicit regularization is essentially ubiquitous in modern machine learning approaches, including stochastic gradient descent for training deep neural networks, and ensemble methods (such as random forests and gradient boosted trees). In explicit regularization, independent of the problem or model, there is always a data term, that corresponds to a likelihood of the measurement, and a regularization term that corresponds to a prior. By combining both using Bayesian statistics, one can compute a posterior, that includes both information sources and therefore stabilizes the estimation process. By trading off both objectives, one chooses to be more aligned to the data or to enforce regularization (to prevent overfitting). There is a whole research branch dealing with all possible regularizations. In practice, one usually tries a specific regularization and then figures out the probability density that corresponds to that regularization to justify the choice. It can also be physically motivated by common sense or intuition. In machine learning, the data term corresponds to the training data and the regularization is either the choice of the model or modifications to the algorithm. It is always intended to reduce the generalization error, i.e. the error score with the trained model on the evaluation set (testing data) and not the training data. One of the earliest uses of regularization is Tikhonov regularization (ridge regression), related to the method of least squares.

>> Section 13

$\text{prox}_{\lambda, R, g}(\mathbf{w}_g) = \begin{cases} (1 - \lambda \|\mathbf{w}_g\|_2) \mathbf{w}_g, & \text{if } \|\mathbf{w}_g\|_2 > \lambda \\ 0, & \text{if } \|\mathbf{w}_g\|_2 \leq \lambda \end{cases}$